

TokTickIT — Lab 2 Test Plan

1. Test Strategy

This test plan is based on the Sprint Engineering Specification, API Spec, and UI Spec before implementation is complete (Test-DD). Testing follows a TDD approach: write the test, implement the required behavior, and keep the test passing.

Testing covers four levels: **Unit, API, UI, and E2E**. Tests are grouped by screen or endpoint and consolidated into the defined test files rather than creating a separate file for each scenario.

Coverage includes:

- Happy paths and successful operations
- Validation and boundary conditions
- Ownership and cross-Requester isolation
- Server and network failure handling
- Loading, empty, and no-results states
- Accessibility and visual requirements
- Attachment upload, download, and soft-removal lifecycle

2. Planned Tests

`server/tests/lab-02/create-ticket.api.test.ts`

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UNIT-01	Unit	BR-01	Ticket Number generator format TKT-<YYYY>-<6-digit seq>	Generated code matches format and is unique per call	Pending
API-01	API	AC-01	POST /create-ticket with valid data	201; one Ticket saved; backend-generated Ticket Number returned	Pass
API-02	API	AC-05	POST /create-ticket with empty summary	400 with fieldErrors for summary ; no Ticket persisted	Pass
API-03	API	AC-06	POST /create-ticket with description < 20 chars	400 naming the 20-char minimum	Pass
API-04	API	AC-07	POST /create-ticket with summary = exactly 150 chars	201; Ticket created (upper boundary passes)	Pass

Test ID	Type	Requirement	What It Tests	Expected Result	Final
API-05	API	AC-08	POST /create-ticket with summary = 151 chars	400; Ticket not created (upper boundary fails)	Pass
API-06	API	AC-14	Ticket create succeeds, Attachment upload then fails	Ticket persists with its number; failed Attachment reported separately (BR-21)	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/server

- ✓ tests/lab-02/create-ticket.api.test.ts (7) 485ms
 - ✓ POST /api/create-ticket (7) 485ms
 - ✓ UNIT-01: should generate a unique ticket number in the correct format 441ms
 - ✓ API-01: should create a ticket with valid data
 - ✓ API-02: should reject a ticket with an empty summary
 - ✓ API-03: should reject a description shorter than 20 characters
 - ✓ API-04: should accept a summary with exactly 150 characters
 - ✓ API-05: should reject a summary with 151 characters
 - ✓ API-06: should keep the ticket when attachment upload fails

Test Files 1 passed (1)
Tests 7 passed (7)
Start at 14:34:58
Duration 7.70s (transform 88ms, setup 0ms, collect 5.89s, tests 485ms, environment 0ms, prepare 878ms)

server/tests/lab-02/attachments.api.test.ts

Test ID	Type	Requirement	What It Tests	Expected Result	Final
Test ID	Requirement	What It Tests	Expected Result		
-----	-----	----- -----	----- -----		
API-07	AC-09 / BR-22	Upload .gif attachment	415; attachment not created		
API-08	AC-10 / BR-23	Upload valid PDF > 5 MB	413; attachment not created		
API-09	AC-11 / BR-24	Upload 6th active attachment	422; attachment not created		
API-10	AC-21 / FR-12	Upload valid attachment to owned ticket	201; Attachment created as ACTIVE		

Test ID	Type	Requirement	What It Tests	Expected Result	Final
API-11	FR-05 / FR-12	List ticket attachments	200; active + removed metadata returned		
API-12	AC-22 / FR-13	Download active attachment	200; original file content returned		
API-13	AC-23 / FR-14	Remove attachment with valid reason	200; status becomes REMOVED		
API-14	BR-26	Remove attachment with empty reason	400; attachment remains ACTIVE		
API-15	BR-27	Download removed attachment	404; file content not returned		
API-16	BR-11 / FR-15	Requester's attachment belongs to another Requester	404; no attachment data revealed		
API-16	BR-24	Add attachment after one is removed	201; removed attachment does not count toward 5-active limit		

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```

✓ tests/lab-02/attachments.api.test.ts (11) 591ms
  ✓ Attachment API (11) 590ms
    ✓ API-07: rejects GIF attachments without creating one
    ✓ API-08: rejects a valid PDF larger than 5 MB
    ✓ API-09: rejects a sixth active attachment
    ✓ API-10: uploads a valid attachment to an owned ticket
    ✓ API-11: lists active and removed attachment metadata
    ✓ API-12: downloads an active attachment with its original content
    ✓ API-13: removes an attachment with a valid reason
    ✓ API-14: rejects an empty removal reason and keeps the attachment active
    ✓ API-15: does not download a removed attachment
    ✓ API-16: hides another requester's attachment
    ✓ API-24: allows an upload after an attachment is removed

```

Test Files 1 passed (1)

Tests 11 passed (11)

Start at 22:56:49

Duration 2.72s (transform 93ms, setup 0ms, collect 1.25s, tests 591ms, environment 0ms, prepare 449ms)

server/tests/lab-02/my-tickets.api.test.ts

Test ID	Type	Requirement	What It Tests	Expected Result	Final
API-16	API	AC-04	GET /api/tickets as Requester B	List contains none of Requester A's Tickets	Pass
API-17	API	AC-15	GET /api/tickets?search=<partial ticket #>	Only Tickets whose number contains the text returned	Pass
API-18	API	AC-16	GET /api/tickets?category=&requestedPriorityId=combined	Only Tickets matching both filters returned (BR-13)	Pass
API-19	API	AC-17	Toggle sortDir on sortBy=createdAt	List order reverses accordingly	Pass
API-20	API	AC-18	Page forward beyond page size	Next set of Tickets loads; page / totalPages metadata correct	Pass
API-21	API	BR-15	page / pageSize with invalid values (e.g. negative, non-numeric)	Falls back to defaults (page 1, size 10) instead of erroring	Pass
API-22	API	AC-25	GET /api/dev-requesters with one inactive Requester seeded	Inactive Requester absent from response (BR-06)	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/server

✓ tests/lab-02/my-tickets.api.test.ts (7)
✓ GET /api/tickets (6)
✓ API-16: should return only tickets owned by requester
✓ API-17: should return only tickets matching the search text
✓ API-18: should return only tickets matching all filters
✓ API-19: should reverse ticket order when sortDir is toggled
✓ API-20: should return the next set of tickets on the next page
✓ API-21: should fall back to default pagination for invalid values
✓ GET /api/dev-requesters (1)
✓ API-22: should exclude inactive requester from the response

Test Files 1 passed (1)

Tests 7 passed (7)

Start at 23:14:16

Duration 766ms (transform 84ms, setup 0ms, collect 242ms, tests 151ms, environment 0ms, prepare 103ms)

server/tests/lab-02/ticket-detail.api.test.ts

Test ID	Type	Requirement	What It Tests	Expected Result	Final
API-23	API	AC-03	GET /api/tickets/:ticketCode for Requester A's Ticket while B is current	404; no Ticket data returned (BR-11)	Pass

RUN v2.1.9 C:/KMUTT/3.1/Software/Pai/server

✓ tests/lab-02/ticket-detail.api.test.ts (2) 312ms
✓ GET /api/tickets/:ticketNumber (2) 310ms
✓ returns the current requester's owned ticket detail
✓ API-23: should return 404 when requester B tries to access requester A's ticket

Test Files 1 passed (1)

Tests 2 passed (2)

Start at 10:27:30

Duration 3.47s (transform 340ms, setup 0ms, collect 1.11s, tests 312ms, environment 1ms, prepare 818ms)

client/tests/lab-02/CreateTicket.test.tsx

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UI-01	UI	AC-02	Navigate to Create Ticket with no Requester selected	Redirected to Requester Selection screen (BR-08)	Pass

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UI-02	UI	AC-05	Click Submit with Summary empty	Inline field message shown; no API call made	Pass
UI-03	UI	AC-06	Type a 19-character Description and submit	Boundary message names the 20-char minimum	Pass
UI-04	UI	AC-12	Click Submit on a valid form	Submit disabled + busy indicator until request resolves (BR-19)	Pass
UI-05	UI	AC-13	Submit valid form while backend is unreachable	Safe error banner shown; all field values remain in the form	Pass
UI-06	UI	AC-28	Tab through the Create Ticket form using keyboard only	Every control reachable in logical order with visible focus indicator	Pass
UI-07	UI	BR-01, BR-03	Render Create Ticket system-generated fields (Ticket #, Date, Requester)	Read-only fields use distinct shading, no focus ring, not tab-stoppable	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

✓ tests/lab-02/CreateTicket.test.tsx (9) 3908ms

✓ Create Ticket screen (9) 3907ms

✓ UI-01: redirects to requester selection without a selected requester

✓ shows the create ticket form when a requester is selected

✓ navigates to create ticket from the top bar

✓ UI-02: shows an inline summary error without calling the create API

✓ UI-03: reports the 20-character description minimum 984ms

✓ UI-04: disables submit while a valid request is pending 1153ms

✓ UI-05: shows a safe error and preserves values when the backend is unreachable 1175ms

✓ UI-06: reaches every create-ticket form control with the keyboard 343ms

✓ UI-07: renders generated fields as shaded, non-focusable read-only fields

Test Files 1 passed (1)

Tests 9 passed (9)

Start at 21:21:14

Duration 5.21s (transform 129ms, setup 69ms, collect 265ms, tests 3.91s, environment 425ms, prepare 123ms)

client/tests/lab-02/MyTickets.test.tsx

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UI-11	UI	AC-19	Open My Tickets for a Requester with zero Tickets	Empty state (not No-Results) shown with Create Ticket CTA	Pass
UI-12	UI	AC-20	Apply filters that match no owned Tickets	No-Results state shown with a Clear Filters action	Pass
UI-13	UI	AC-26	Use Change Requester to pick a different active Requester	My Tickets reloads showing only the new Requester's Tickets (BR-07)	Pass

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✓ tests/lab-02/MyTickets.test.tsx (3) 683ms

✓ My Tickets screen (3) 682ms

✓ UI-11: shows the empty state and Create Ticket CTA for a requester with no tickets

✓ UI-12: shows no-results state and clears active filters

✓ UI-13: changes requester and reloads My Tickets with the new requester

Test Files 1 passed (1)

Tests 3 passed (3)

Start at 21:31:44

Duration 3.54s (transform 260ms, setup 115ms, collect 556ms, tests 683ms, environment 839ms, prepare 431ms)

client/tests/lab-02/AttachmentSection.test.tsx

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UI-08	UI	AC-09, AC-10	Select a .gif or an oversized PDF in the Attachment control	Rejected client-side before any upload call, with a clear message	Pass
UI-09	UI	AC-11	Attachment control on a Ticket with 5 active Attachments	Remaining-slots indicator shows 0; a 6th file is blocked with a limit message	Pass
UI-10	UI	AC-23	Click Remove on an Attachment without entering a reason	Removal blocked until a non-empty reason is entered	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

- ✓ tests/lab-02/AttachmentSection.test.tsx (3)
 - ✓ Attachment controls (3)
 - ✓ UI-08: rejects gif or oversized PDF files before upload
 - ✓ UI-09: shows remaining slots and blocks the 6th file
 - ✓ UI-10: blocks removal without a non-empty reason

Test Files 1 passed (1)

Tests 3 passed (3)

Start at 22:55:28

Duration 1.45s (transform 80ms, setup 92ms, collect 188ms, tests 277ms, environment 418ms, prepare 132ms)

client/tests/lab-02/RequesterTicketDetail.test.tsx

Test ID	Type	Requirement	What It Tests	Expected Result	Final
UI-15	UI	AC-21	Add a valid Attachment from Ticket Detail	New Attachment appears in list as Active without a page reload	Pass
UI-16	UI	AC-24	View Ticket Detail after an Attachment was soft-removed	Attachment shown greyed-out with metadata/reason; download control disabled	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

- ✓ tests/lab-02/RequesterTicketDetail.test.tsx (2)
 - ✓ Requester ticket detail attachments (2)
 - ✓ UI-15: adds a valid attachment from ticket detail without reload
 - ✓ UI-16: renders a removed attachment as greyed-out with reason and disabled download

Test Files 1 passed (1)

Tests 2 passed (2)

Start at 22:45:53

Duration 1.30s (transform 136ms, setup 60ms, collect 273ms, tests 203ms, environment 365ms, prepare 146ms)

e2e/lab-02/requester-ticket-flow.spec.ts

Test ID	Type	Requirement	What It Tests	Expected Result	Final
E2E-01	E2E	AC-01	Complete Requester selection → Create Ticket submission flow with valid data	Ticket is created successfully, the backend-generated Ticket Number is displayed, and Ticket Details opens	Pass

Test ID	Type	Requirement	What It Tests	Expected Result	Final
E2E-02	E2E	AC-03, AC-04	Requester A creates a Ticket; switch to Requester B and search My Tickets / open A's Ticket by number	Requester B never sees A's Ticket in the My Tickets list or Ticket Detail	Pass
E2E-03	E2E	AC-21, AC-22, AC-23, AC-24	Full Attachment lifecycle: add, download, soft-remove with reason	Attachment appears as Active, can be downloaded, then shows as Removed with metadata and reason; download is disabled after removal	Pass
E2E-04	E2E	AC-25, AC-26	Open Requester selector with an inactive Requester seeded, select an active Requester, then Change Requester	Inactive Requester is absent; switching Requester reloads My Tickets with only the newly selected Requester's Tickets	Pass
E2E-05	E2E	—	Opens an invalid Ticket and verifies the application's validation/error handling	Invalid Ticket is rejected and an appropriate error state is shown without exposing Ticket data	Pass
E2E-06	E2E	AC-15, AC-16	Search My Tickets by partial Ticket Number and apply Category + Requested Priority filters	Only Tickets matching the search text and/or all selected filters are displayed	Pass
E2E-07	E2E	AC-11	Adds Attachments up to the active limit, removes one, then attempts to fill the available slot again	Attachment limit is enforced; after one Attachment is removed, the available slot can be used again	Pass
E2E-08	E2E	AC-16	Applies a Category and Requested Priority filter without entering a search term	My Tickets displays only Tickets matching the applied filters; unrelated Tickets are excluded	Pass
E2E-09	E2E	AC-11	Attempts to exceed the maximum number of Attachments on Create Ticket	Upload beyond the configured Attachment limit is rejected and the limit message is shown	Pass

```
npm notice run software-engineer@1.0.0 test:e2e
npm notice run playwright test e2e/lab-02/requester-ticket-flow.spec.ts
[WebServer] npm notice run toktickit-client@1.0.0 dev
[WebServer] npm notice run vite --host 127.0.0.1
```

Running 9 tests using 1 worker

```
✓ 1 ...ket-flow.spec.ts:48:6 › Requester ticket flow › E2E-01: creates a ticket and shows the backend-generated
✓ 2 ...er-ticket-flow.spec.ts:54:6 › Requester ticket flow › E2E-02: prevents another requester from seeing the
✓ 3 ...lab-02\requester-ticket-flow.spec.ts:71:6 › Requester ticket flow › E2E-03: completes the attachment list
✓ 4 ...uester-ticket-flow.spec.ts:95:6 › Requester ticket flow › E2E-05: shows validation errors for an invalid
✓ 5 ...2e\lab-02\requester-ticket-flow.spec.ts:111:6 › Requester ticket flow › E2E-06: searches and filters My
✓ 6 ...cket-flow.spec.ts:130:6 › Requester ticket flow › E2E-07: refills the sixth attachment slot after deletion
✓ 7 ...ter-ticket-flow.spec.ts:158:6 › Requester ticket flow › E2E-08: captures My Tickets with a filter and no
✓ 8 ...r-ticket-flow.spec.ts:175:6 › Requester ticket flow › E2E-09: captures the seventh file limit on Create
✓ 9 ...-flow.spec.ts:191:6 › Requester ticket flow › E2E-04: switches requester data and excludes inactive requesters
```

9 passed (15.9s)

3. Traceability Summary

- The full Attachment lifecycle is tested at API, UI, and E2E levels.
- Responsive behavior is tested for both Create Ticket and My Tickets.
- Server tests use 4 files; client tests use 4 files; all E2E and responsive tests use `requester-ticket-flow.spec.ts`.
- All tests start as `Pending` and are updated to `Pass` or `Fail` after implementation and execution.